Computer assignment

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Question 1

In a gambling game, a player wins the game if they roll 10 fair, six-sided dice, and get a sum of at least 40.

Approximate the probability of winning by simulating the game 10^4 times.

```
set.seed(2)
rolls = sample(x=1:6, size=10^4*10, replace=TRUE)
result = matrix(rolls, nrow=10^4, ncol=10)
rollsums = apply(result, 1, sum)
#prob of winning
mean(rollsums>=40)
```

```
## [1] 0.2066
```

Compute the Central Limit Theorem approximation $P(Y \geq 40)$, where Y is the sum of 10 dice, and compare it to the Monte Carlo approximation obtained above. You can use the fact that $E(Y) = 35$, $Var(Y) = 175/6$.

```
z=(40-35)/sqrt(175/6)
1-pnorm(z)
```

```
## [1] 0.1772697
```

```
mean(rollsums>=40) - (1-pnorm(z))
```

```
## [1] 0.02933026
```

The result from CLT is not that close to the Monte Carlo approximation.

Question 2

The frequency table given in the sheet summarises 320 counts modelled as values taken by i.i.d. random variables with common $\mathrm{Bin}(4, p)$ distribution, i.e.

$$P(X = x) = \binom{4}{x} p^x (1-p)^{4-x}, \quad \text{for } x = 0, 1, 2, 3, 4$$

for some unknown p.

Recall $E(X) = np$, estimate p using the method of moments.

```
x=0:4
freq=c(130,133,49,7,1)
empirical.mean=sum(x*freq)/sum(freq)
phat=empirical.mean/4
phat
```

```
## [1] 0.2
```

Using the previous point, find expected frequencies (E) for each of the classes ‘0’, ‘1’, ‘2’, ‘3’ and ‘4’. Round to the nearest integer.

```
E=round(dbinom(x,4,phat)*sum(freq))
E
```

```
## [1] 131 131  49   8   1
```

Compute standardised residuals (SR) given by $SR = \frac{O-E}{\sqrt{E}}$ for each of the classes ‘0’, ‘1’, ‘2’, ‘3’ and ‘4’, where O represents the observed frequencies. If $|SR| < 2,$ then the fitted binomial model is said to be a good model for the data. Comment on the goodness of fit.

```
SR=abs((freq-E)/sqrt(E))
SR
```

```
## [1] 0.08737041 0.17474081 0.00000000 0.35355339 0.00000000
```

```
mean(SR)
```

```
## [1] 0.1231329
```

All the SR values are small, indicating a good fit.

Question 3

Generate a random sample of size 25 (using a seed equal to 100) from a normal distribution with mean $\mu = 3$ and standard deviation $\sigma = 1.5$. Assume σ is known and we want to estimate μ. Using the sample generated, find a 95% confidence interval (CI) for μ.

```
set.seed(100)
n=25
sigma=1.5
x=rnorm(25, 3, sigma)
z=-qnorm(0.025)
lb=mean(x)-z*sigma/sqrt(n)
ub=mean(x)+z*sigma/sqrt(n)
c(lb, ub)
```

```
## [1] 2.574269 3.750247
```

Repeat the process in (a) 20 times. Using your 20 samples, calculate 20 CIs for μ. How many of these 20 intervals contain the true mean $\mu = 3$? Output this number from your code, but no need to print the 20 CIs themselves.

```
set.seed(100)
xmat=matrix(rnorm(n*20, 3, sigma),nrow=20, ncol=n)
xmean=apply(xmat,1,mean)
xlb=xmean-z*sigma/sqrt(n)
xub=xmean+z*sigma/sqrt(n)

sum((xlb<3) & (xub>3))
```

```
## [1] 19
```

19 of these intervals contain the true mean.

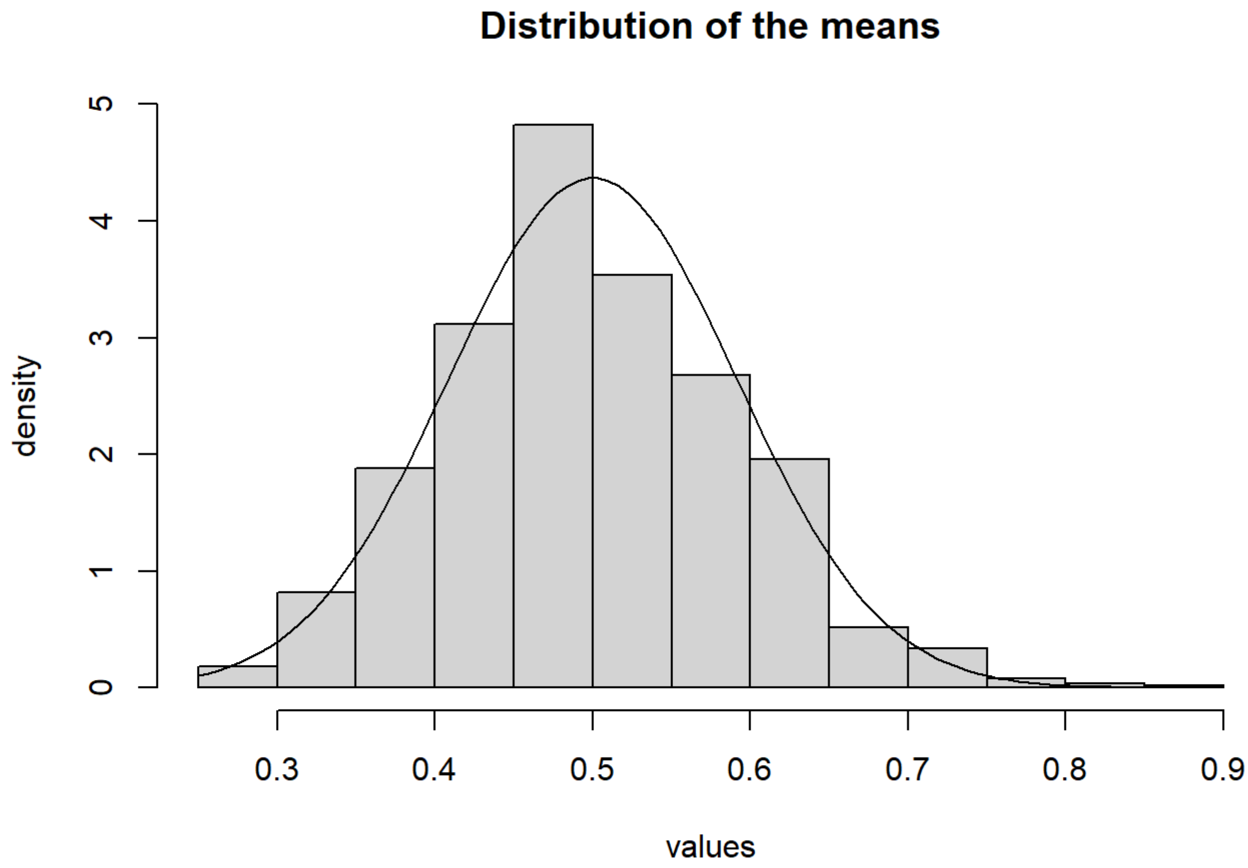
Question 4

Generate a random sample of size 30 from the exponential distribution with parameter $\lambda = 2$ and find the mean of your sample. Repeat this process 1000 times and draw a histogram of these 1000 means.

```
set.seed(100)
data=rexp(30,2)
mean(data)
```

```
## [1] 0.368984
```

```
xmat=matrix(rexp(30*1000, 2), nrow=1000, ncol=30)
xmean=apply(xmat, 1, mean)
hist(xmean, prob=T, main = 'Distribution of the means', xlab="values", ylab="density")
curve(dnorm(x, mean=1/2, sd=sqrt(1/(4*30))), add=T)
```



The fit seems reasonable here with a small amount of bias.